



Instructor: Date:

Learner: Regional Branch:

Program: Center:

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**Let's Start!**

An Approach to Software Development

Object Oriented Programming

Learning Outcome:

Identify and explain the characteristics of software.

Discuss information about designing principles in software programming

Ask your partner:

- Why do you think object oriented programming is important?
- Can you think of an example that shows what an object in programming is?

Can Do Statements:

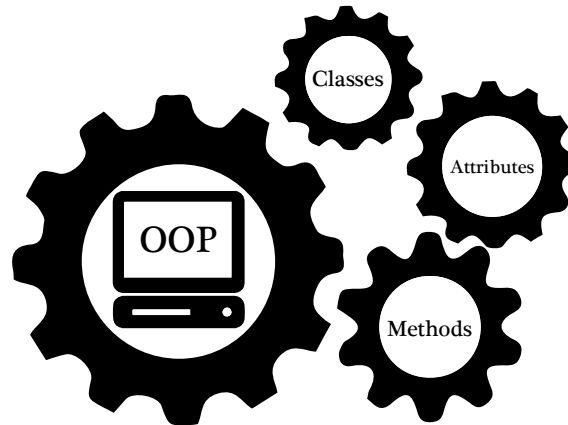
- I can talk about the concept of object oriented programming.
- I can talk about objects as components of a program.
- I can use an everyday example to talk about object oriented programming.





Let's Explore!

These students are studying very hard for the quiz about object oriented programming.



Instructions: Read the text below.

Sara: Juan, what are you doing? Are you studying for the quiz?

Juan: No, I am not studying. What are you talking about? What quiz?

Sara: The final one. Laura and I are reviewing our notes about object oriented programming. We aren't checking complex ideas to understand the concept. Instead, we're trying to link its definitions with everyday examples and simple general concepts.

Juan: Can you explain that to me?

Sara: Yes! Basically, object oriented programming is an approach to software development, which uses objects to solve different problems.

Juan: Objects?

Sara: Objects are its basic units. They are components that compose the general performance of the program. It means that the object knows how the actions of any program work and how they interact with each other.

Juan: Any example?

Sara: Sure. Because we aren't studying from books, let's think about an everyday example. Mmm, a person! What is the first thing you expect a person to have?

Juan: A name, right?

Sara: Good! A name is to a person what objects are to OOP; That is the property of an object. In addition, you can also expect the person to have a certain skill or ability, such as writing or singing; that is, in terms of software programming, the functionalities or methods that an object has. Now, to create codes around those objects, you need to have in mind some steps: First, we need to understand the components of the object we want to create; second, we need to describe them, and finally, you need to bring them together so they can interact and fulfill the result you want. For example, I'm creating a program whose function is to lead a person who walks from point A to point B. First, you describe the way and the person separately; next, you include the methods, and to conclude, you join them together so the person can walk on that road from A to B.





Let's Practice!

Basic Concepts on OOP

Instructions: Match the sentences from Column A with Column B.

COLUMN A

1. You are not
2. Juan isn't
3. If you don't use objects to solve problems in software development,

COLUMN B

- a. you are not applying an object oriented programming.
- b. the concept of "objects", which are the basic unit of an OOP.
- c. reviewing the notes about the concept of object oriented programming.



Let's Practice!

Everyday Examples of an OOP

Instructions: Read the sentences. Write the missing word(s) in the blank. Use: property, object, abilities.

1. The name of a person is the first _____ you expect it to have.
2. Writing or singing are skills which are the examples of an _____ in software.
3. A person can have different _____ to do something, those are the methods of that person.





Extension Activity

Concepts of Object Oriented Programming

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Go to the nearest library or search on the Internet for additional information on OOP.
- Design a brochure about advances on OOP.
- Share your brochure to the class.

Self-Assessment

Yes

No

- I read information about action planning guides
- I design a brochure showing advances on OOP.
- I share my brochure to the class.

☐	☐
☐	☐
☐	☐



Target Words

- review** (verb): to re-examine and revise
e.g. Laura and I are reviewing our notes about object oriented programming.
- development** (noun): the process of a certain project
e.g. Basically, object oriented programming is an approach to software development, which uses objects to solve different problems.
- solve** (verb): to find a solution to a problem
e.g. Object oriented programming uses objects to solve different problems.
- work** (verb): to operate
e.g. It means that the object knows how the actions of the program work and how the interactions with other components are.
- fulfill** (verb): to accomplish a duty or task
e.g. You need to bring them together so they can interact and fulfill the result you want.
- bring something together** (verb): to join two or more things
e.g. Bring them together so they can interact and fulfill the result you want.



Learning Points

Present Continuous Tense in Negative Statements

The **Present Continuous** is used to talk about actions that happen at the moment of speaking.

To express that an action is taking place now, follow the next structure:

Subject + **verb to be in present + verb -ing** + complement

For example:

*I **am reading** about programming.*

*They **are talking** about object oriented programming.*

*She **is trying** to understand what programming is.*

If you want to formulate a negative statement you just have to add the negative form of the **verb to be** and keep the rest of the structure:

Subject + **verb to be in negative form + verb-ing** + complement

For example:

*I **am not reading** about programming.*

*They **are not talking** about object oriented programming.*

*She **is not trying** to understand what programming is.*

Examples

- *No, I am not studying. What are you talking about?*
- *We aren't checking complex ideas to understand the concept.*
- *Because we aren't studying from books, let's think about a general and simple example.*





Learning Points

Sequence Words

You use **sequence words** to link two or more consecutive ideas in a paragraph.

We need to use sequence words to understand the order in which an event takes place. Look at the following chart; it contains three general parts of a sequence and some of the words that can introduce them.

Beginning	Continuing	Ending
<i>First,</i>	<i>Second,</i>	<i>Finally,</i>
<i>First of all,</i>	<i>Later,</i>	<i>To conclude,</i>
<i>To begin with,</i>	<i>Next,</i>	<i>In the end,</i>

For example:

*If you want to understand about object oriented programming you need to follow the next process. **First**, read and understand what type of approach it is. **Next**, read about classes and objects, and their importance in this programming. **Finally**, you can try to start solving computation problems to find solutions to a problem.*

Examples

- First, you describe the car and the person separately; next, you include the methods, and to conclude, you join them together so the person can drive the car from A to B.*

